

Sampling Distribution Notation

	Parameters	Statistics	Sampling Distribution Mean	Sampling Distribution Standard Deviation
Proportions: Categorical Data				
Means: Quantitative Data				

Sampling Distribution Conditions for Sample Means

Random - As long as the sampling method is _____, our mean is an unbiased estimator.

Independent - When sampling, we have to make sure the _____ is satisfied.

Normal - No LARGE COUNTS!

Central Limit Theorem:

- The central limit theorem (CLT) states that when the sample size is sufficiently large, a sampling distribution of the mean of a random variable will be _____.
- The central limit theorem requires that the sample values are _____ of each other and that n is _____.

Therefore....

- If the population distribution is _____, then so is the sampling distribution of $\bar{x}$. This is true no matter what the sample size n is.
- If the population distribution is _____ (or has an unknown shape), the central limit theorem tells us that the sampling distribution of $\bar{x}$ will be approximately Normal in most cases if $n \geq 30$.

Sampling Distribution for a Sample Mean

Shape	Center	Spread

As long as...	As long as...	As long as...
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Example Problems for a Sampling Distribution for a Sample Mean

Suppose the heights of young women are normally distributed with $\mu = 64.5$ inches and $\sigma = 2.5$ inches. What is the probability that the mean height of an SRS of 10 young women is greater than 65 inches?		
Step 1	Communicate the parameter in the context of the problem. Write down the probability you are trying to find.	
Step 2	Create your sampling distribution by finding the center, spread, and shape. Make sure to check each condition first.	
Step 3	Calculate the probability by converting your data to a z-score and using technology to find the probability. <i>*Drawing the curve is optional*</i>	

Step 4	Conclude the problem by answering the question in a complete sentence with the context of the problem.	
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Example) A particular restaurant (that shall remain nameless) is suspected of undercooking its burgers. The restaurant claims that its burgers are cooked to a medium doneness, with an average internal temperature of 160 degrees Fahrenheit and a standard deviation of 5 degrees Fahrenheit. For the questions that follow, assume that the restaurant's claim is accurate and that the distribution of internal temperatures follows a normal distribution.

(a) What is the probability that a single randomly selected burger is cooked to an internal temperature of 158 degrees or less?

(b) A secret shopper comes in at random times throughout the day to test the internal temperature of the burgers. They select an SRS of 25 burgers and calculate the sample mean. What are the mean and standard deviation of the resulting sampling distribution?

(c) The secret shopper in part (b) obtains a sample mean of $\bar{x} = 158$ degrees Fahrenheit. What is the probability that a random sample of 25 burgers produces a sample mean amount of 158 degrees or less? (*Assume all conditions are met*)

(d) Why is there a difference between the results in (a) and (c)? Explain why one is more likely to happen than the other.